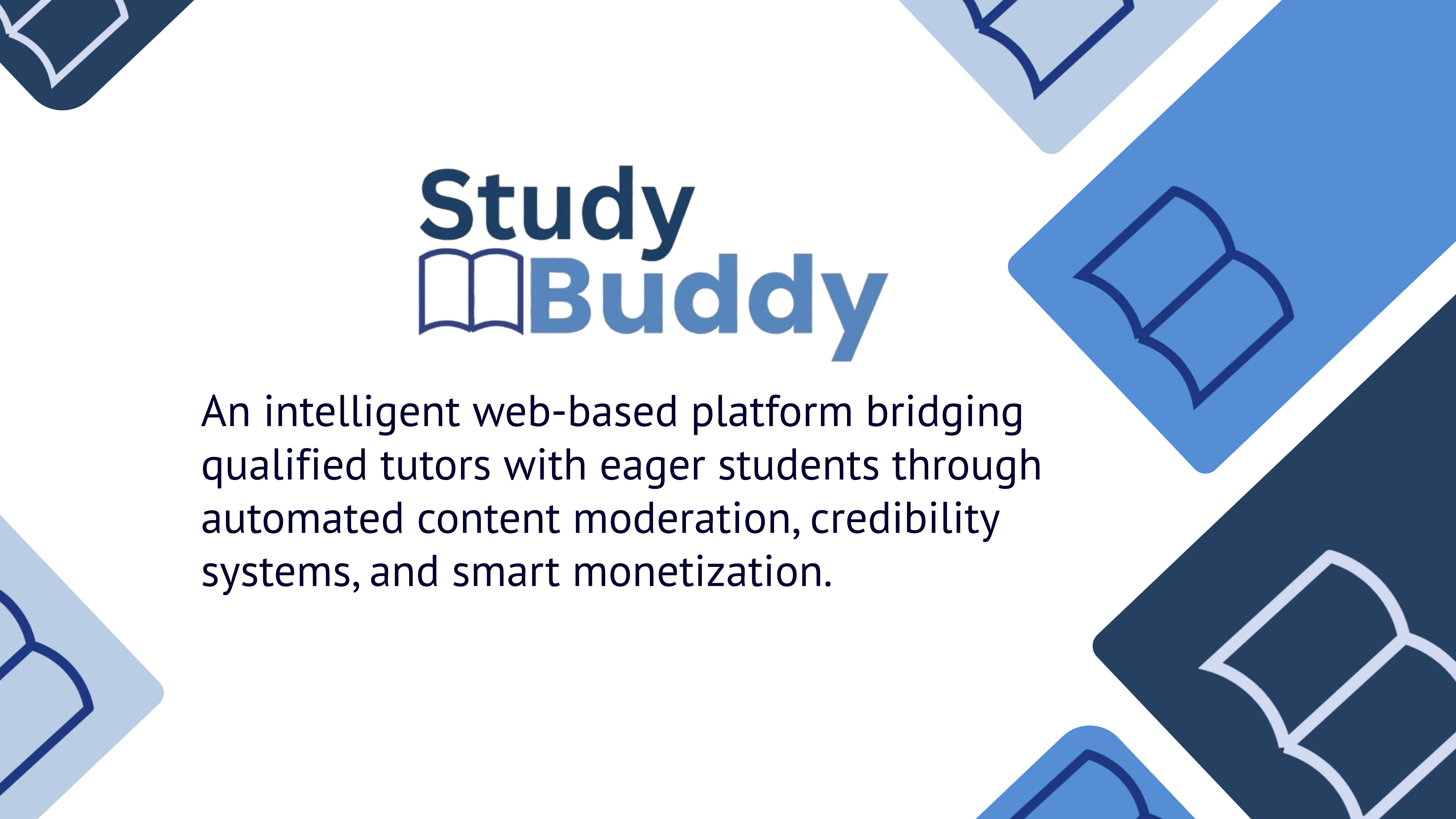
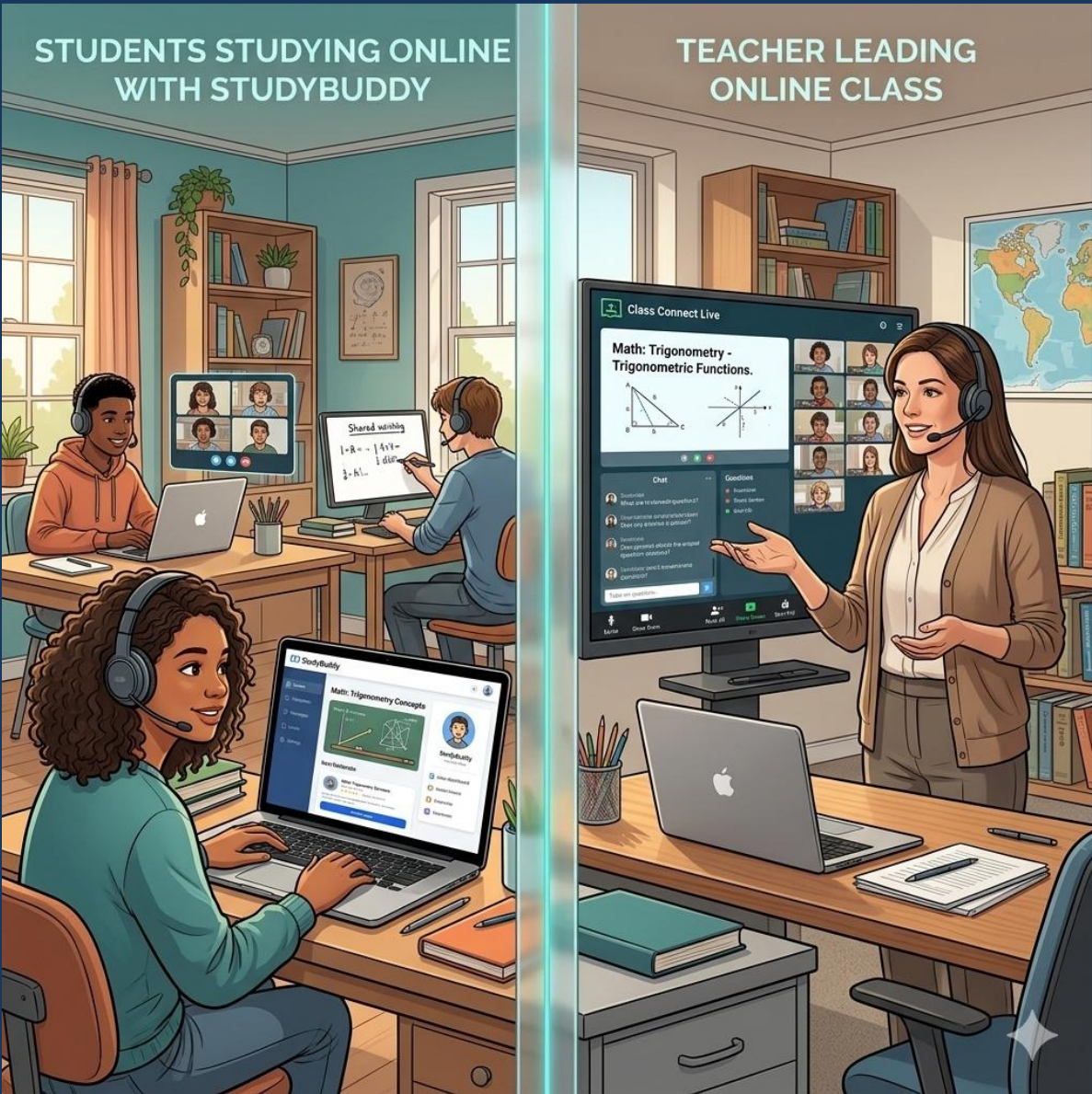


Study Buddy

The background of the slide features several stylized book icons in the corners. In the top-left, there is a dark blue icon. In the top-right, a light blue icon. In the middle-right, a medium blue icon. In the bottom-left, a light blue icon. In the bottom-right, a dark blue icon. The books are depicted with simple outlines and some internal shading to suggest pages.

Study Buddy

An intelligent web-based platform bridging qualified tutors with eager students through automated content moderation, credibility systems, and smart monetization.



What is StudyBuddy?

Introduction:

An Intelligent Ed-Tech Platform



Web-Based Platform

Intelligent platform connecting tutors & students(Groups) globally



Content Moderation

Automated multi-layer pipeline for safe learning



Tutor Credibility

Real student reviews drive tutor trustworthiness



Smart Monetization

Flexible, fair model benefiting tutors & students



Our Promise: Accessible · Safe · Trustworthy

Our TEAM

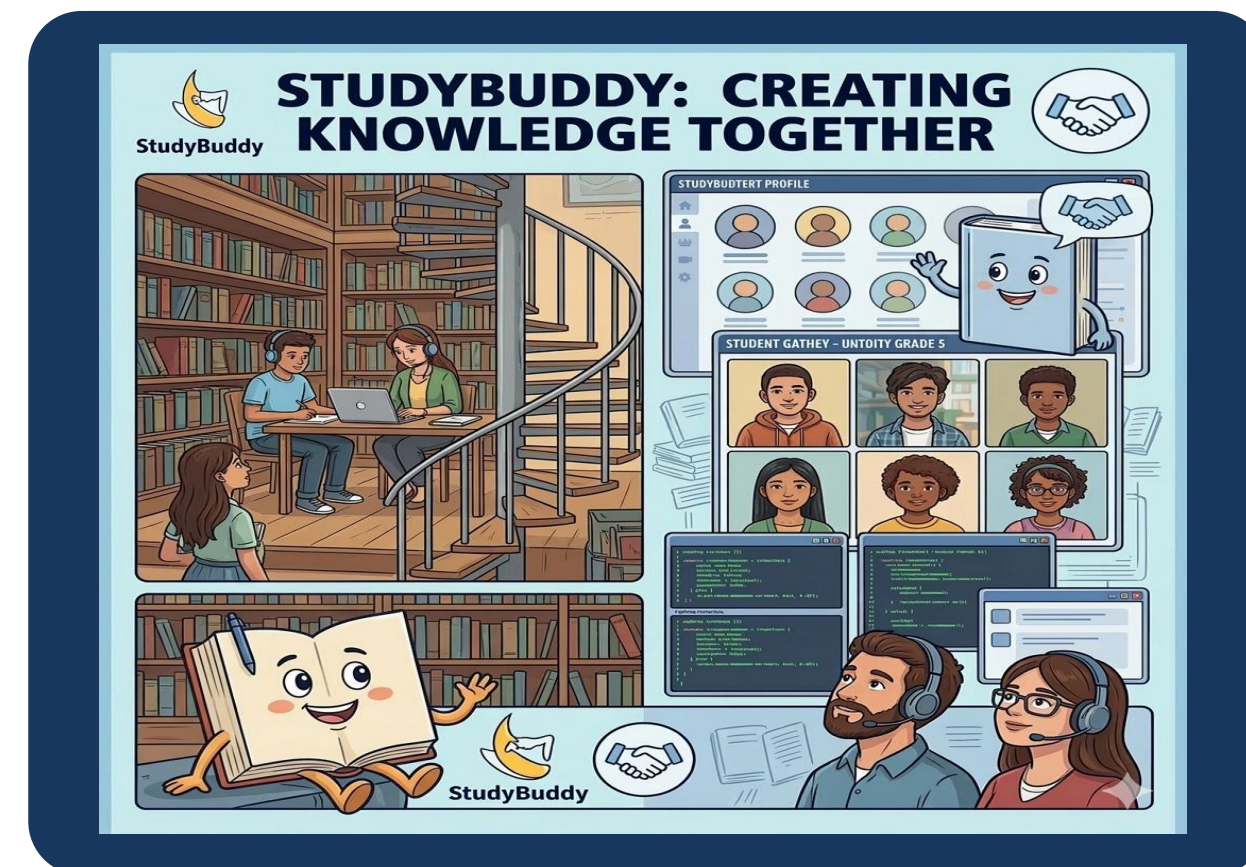
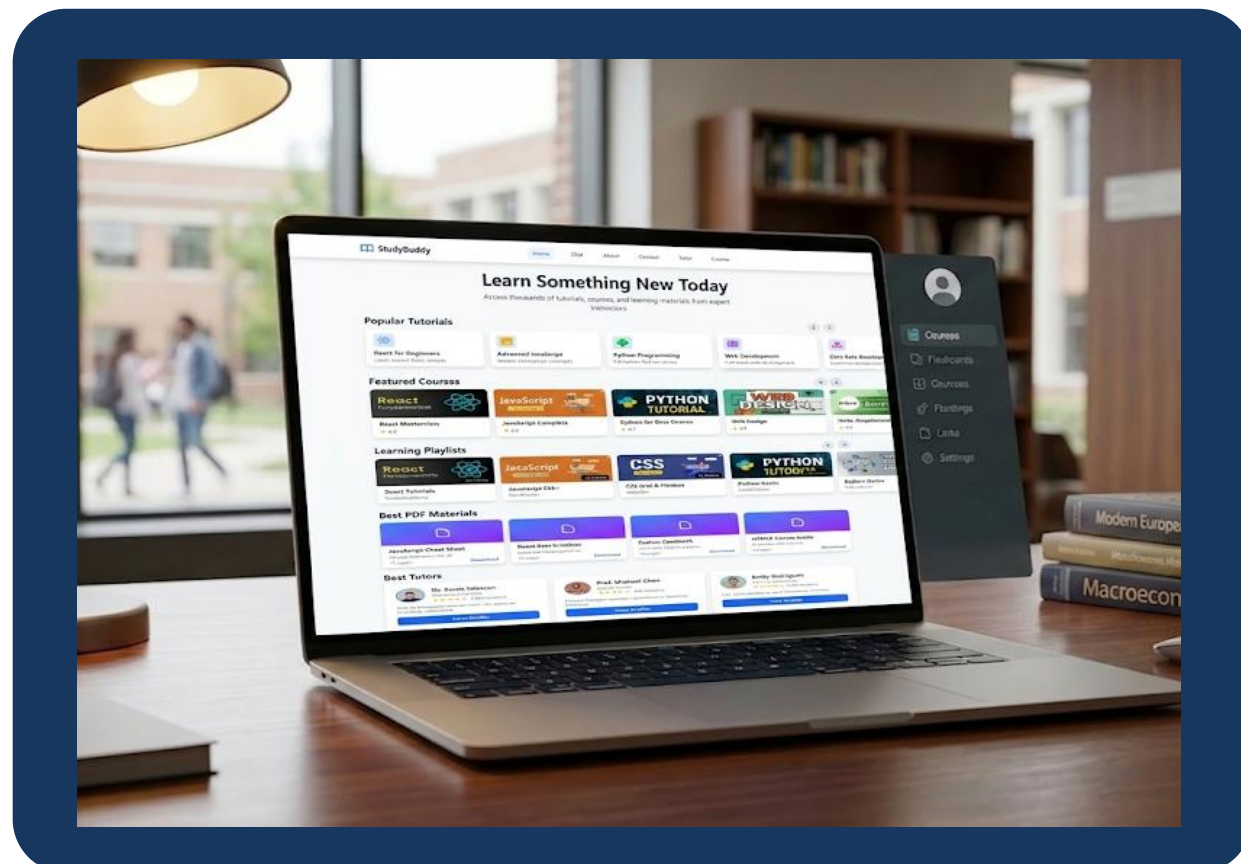
- Pandya Jay
- Chudasama Chandradeep
- Rajyguru Chit
- Hindocha Madhav



Project Profile

Study Buddy revolutionizes online education by creating a trusted ecosystem where verified tutors share structured learning resources whilst maintaining rigorous content safety standards.

Every uploaded resource — PDFs, images, PPTs, audio files, and YouTube video links — passes through an 6-step moderation pipeline, ensuring only appropriate, high-quality content reaches students.



Project Profile

Domain

Education Technology
(Ed-Tech)

Purpose

Connecting Verified Tutors with
students a safe , moderated
platform

Targeted Users

Tutors ,Student ,Admin

Frontend – React.js

Backend – Express.js

Database– MySQL

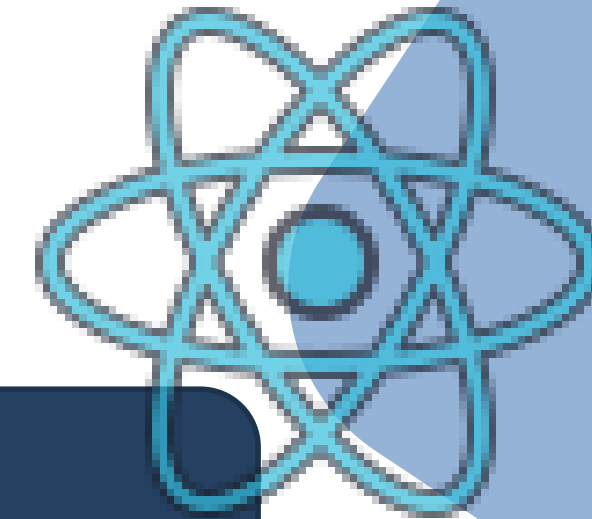
External APIs

Google Vision ,OpenAI ,YouTube
,Data API ,Pytesseract

Technology Stack

Frontend

React.js for responsive, interactive user interfaces



Backend

Express.js (Node.js) for robust server-side operations



Database

MySQL for structured data management



APIs

Google Vision, OpenAI Moderation, YouTube Data API v3, pytesseract



Core Platform Modules



Authentication Module

- Student, Tutor & Admin role assignment
- Secure bcrypt password storage
- Email verification on registration
- Suspend / ban / reactivate controls



Tutor Verification Module

- Mandatory 5-min trial video upload
- Supporting documents submission
- Admin approve / reject workflow
- Tutor profile & skills created on approval



Resource Upload & Moderation

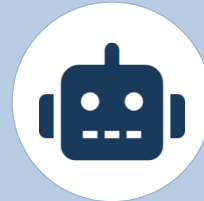
- Supports PDF, image & YouTube video
- Automated 6-step moderation pipeline
- OCR text extraction & category matching
- Random scene analysis for videos

Core Platform Modules



Content Access Module

- Browse free & locked resources
- Tutors sell bundles & test papers
- Students pay to unlock premium content
- Curated playlists per learning need



AI Chat & Group Study Module

- Limited free AI messages for doubt solving
- Upgrade option for more AI interactions
- Group study rooms with free basic access
- Premium moderated collaborative experience

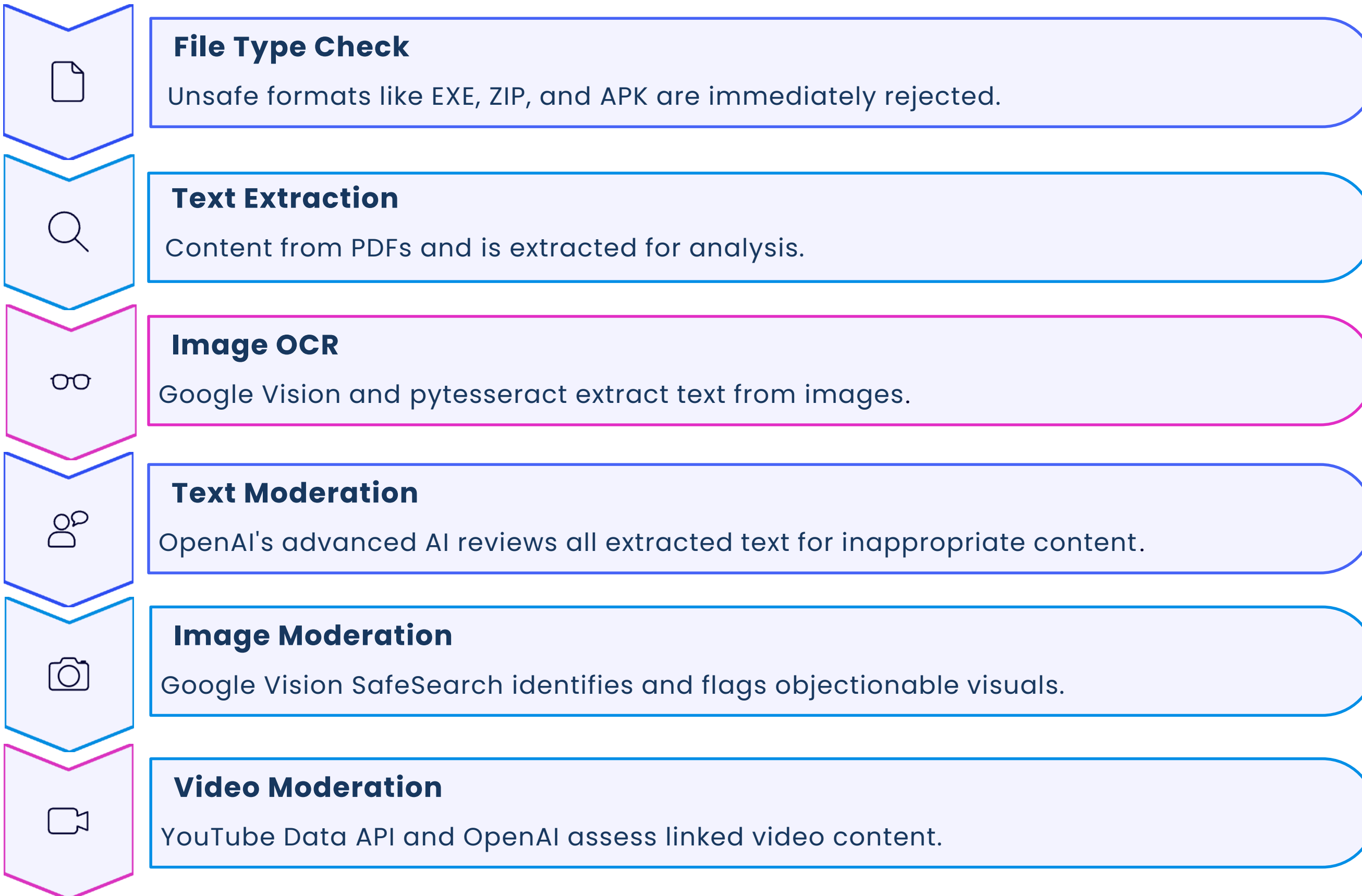


Admin Module

- Central dashboard for platform oversight
- Approve / reject tutor applications
- Manually review automation-flagged content
- Monitor overall platform activity & quality

6-Step Moderation Pipeline

Every resource uploaded undergoes a rigorous, automated pipeline to guarantee safety and quality for students.



Tutor Functionalities



Register & Verify

Submit details and record mandatory 5-minute trial video for student reviews and admin approval.



Upload Resources

Share PDFs, images, or YouTube links directly to the platform.



Organize Content

Create playlists, bundles, and structured learning paths for better student engagement.



Earn Revenue

Track earnings, student engagement, and monetise content through sales and promotional listings.

Student Functionalities



Browse & Explore

Watch tutor trial videos and explore curated learning resources



Review & Earn

Submit mandatory reviews and earn rewards for participation



Access Content

Use free resources or purchase locked content, bundles, test papers



AI Assistance

Chat with AI for doubt solving (limited free, paid for extended access)



Study Together

Join group study rooms (basic free, premium features unlockable)

Admin Capabilities



Approve/Reject Tutors

Review tutor applications and evaluate trial videos to maintain high teaching standards.



Review Flagged Content

Provide manual oversight of automated moderation flags to ensure platform safety and quality.



Manage Platform Users

Oversee user accounts, manage access levels, and resolve user-related issues or disputes.



Monitor Analytics

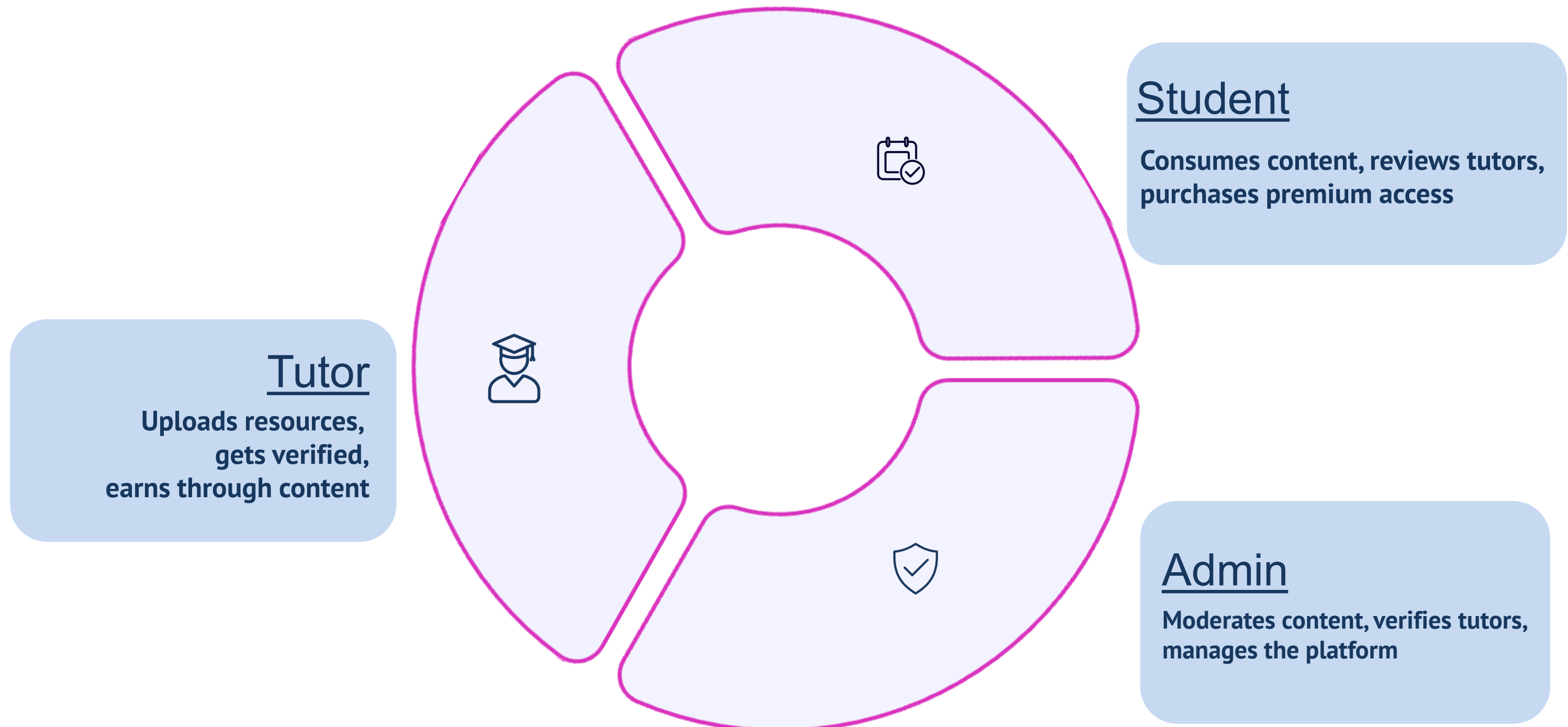
Track platform activity, user engagement, and performance metrics to drive data-informed decisions.



Maintain Quality Standards

Ensure consistent quality across all learning resources and control featured promotional content.

User Ecosystem Architecture



GUI Design (Mock Screen)

User Side

SB StudyBuddy

Search courses, resources...

John Doe

Dashboard

Courses

Resources

Bundles

AI Chat

Group Chat

My Purchases

Demo Mode - Switch Role:

Student

Tutor

Admin

Welcome back, John! 🙌

Ready to continue your learning journey?

Courses Enrolled

5

+2 this month

Resources Purchased

12

Bundles Owned

2

AI Study Buddy

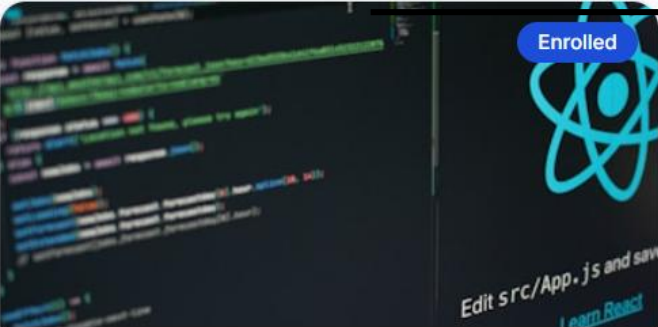
Ask me anything!

My Enrolled Courses

Continue where you left off

View All ↗

Enrolled



Advanced React Patterns and Best Practices

by Sarah Johnson

★ 4.8

\$49.99

Continue Learning

Enrolled



Full Stack Web Development Bootcamp

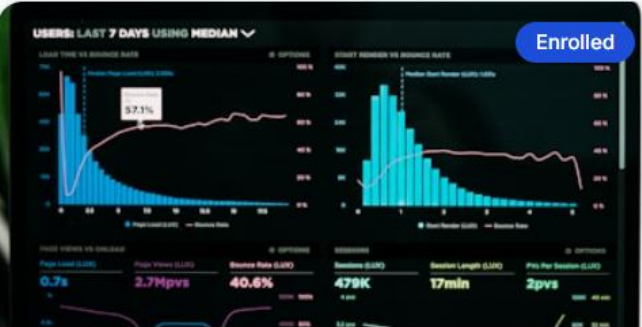
by Michael Chen

★ 4.9

\$79.99

Continue Learning

Enrolled



Data Science with Python

by Emily Rodríguez

★ 4.7

\$59.99

Continue Learning

Browse Courses

Resources

Bundles

Group Chat

StudyBuddy v1.0

Dashboard

Courses

Resources

Bundles

AI Chat

Group Chat

My Purchases

Demo Mode - Switch Role:

Student

Tutor

Admin

Explore Courses

Discover and enroll in courses from expert tutors

Search for courses...

Filters

All

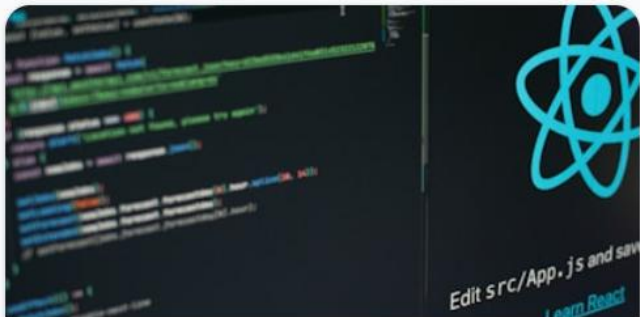
Web Development

Data Science

Design

Business

Marketing



Advanced React Patterns and Best Practices

by Sarah Johnson

★ 4.8

\$49.99

Enroll Now



Full Stack Web Development Bootcamp

by Michael Chen

★ 4.9

\$79.99

Enroll Now



Data Science with Python

by Emily Rodriguez

★ 4.7

\$59.99

Enroll Now



UI/UX Design Fundamentals

by Alex Turner

★ 4.6

\$39.99

Enroll Now



Machine Learning A-Z

by Dr. James Wilson

★ 4.9

\$89.99

Enroll Now



Complete JavaScript Course

by Lisa Anderson

★ 4.8

Free

Enroll Now

Previous

Page 1 of 5

Next

 Dashboard

 Courses

 Resources

 Bundles

 AI Chat

 Group Chat

 My Purchases

Demo Mode - Switch Role:

Student

Tutor

Admin

Learning Resources

Browse standalone resources from expert tutors

Search resources...

 Filters

All

PDF

Image

YouTube



React Hooks Cheat Sheet

by Sarah Johnson

\$9.99

Buy



UI Design Principles Guide

by Alex Turner

Free

View



JavaScript ES6+ Features

by Michael Chen

\$14.99

Buy



Component Architecture Diagram

by Sarah Johnson

\$4.99

Buy



Database Design Patterns

by Dr. James Wilson

\$19.99

Buy



CSS Grid Layout Tutorial

by Lisa Anderson

Free

View

Previous

Page 1 of 4

Next

 Dashboard

 Courses

 Resources

 Bundles

 AI Chat

 Group Chat

 My Purchases

Demo Mode - Switch Role:

Student

Tutor

Admin

Resource Bundles

Get multiple resources at discounted bundle prices

Complete Web Development Bundle

by Sarah Johnson

15 resources

\$129.99

Buy Bundle

Data Science Essentials

by Dr. James Wilson

22 resources

\$199.99

Buy Bundle

UI/UX Design Master Pack

by Alex Turner

18 resources

\$149.99

Buy Bundle

JavaScript Deep Dive Bundle

by Michael Chen

12 resources

\$89.99

Buy Bundle

Python Programming Complete

by Emily Rodriguez

20 resources

\$169.99

Buy Bundle

Full Stack Developer Pack

by Lisa Anderson

25 resources

\$249.99

Buy Bundle

Previous

Page 1 of 3

Next



Demo Mode - Switch Role:

Student

Tutor

Admin

Dashboard

Courses

Resources

Bundles

AI Chat

Group Chat

My Purchases

AI Study Buddy

Daily AI Quota

15 / 50



Hello! I'm your AI Study Buddy. How can I help you today?

12:53 PM

Ask me anything...

Send

Demo Mode - Switch Role:

Student

Tutor

Admin

Dashboard

Courses

Resources

Bundles

AI Chat

Group Chat

My Purchases

General
124 members



Channels +

General 124

Study Group 45

Web Development 67

Data Science 89

Design Discussion 34



Sarah Johnson 1h ago

Hey everyone! Has anyone started the React assignment yet?



Michael Chen 58m ago

Yes! I finished it yesterday. The hooks section was tricky.



Emma Wilson 56m ago

I got stuck on useEffect. Can someone help me understand the cleanup function?

You 55m ago



Sure! The cleanup function runs before the component unmounts or before the effect runs again.



David Kim 53m ago

Also, check out the official React docs. They have great examples!



Type a message...

Send

Admin Side

Dashboard

Tutor Verification

Content Moderation

User Management

Refunds

Demo Mode - Switch Role:

Student

Tutor

Admin

Admin Dashboard

Monitor and manage the StudyBuddy platform

Pending Verifications

12

Flagged Resources

8

Pending Refunds

5

Total Users

15,234

Recent Activity



New tutor application
Sarah Johnson

5 minutes ago



Resource flagged
System

15 minutes ago



Refund request
John Doe

1 hour ago



New user registered
Alice Cooper

2 hours ago



Review Verifications



Moderate Content



Manage Users

Tutor Side

SB

StudyBuddy

🔍

Search courses, resources...

🔔

John Doe

▼

Dashboard

Profile

My Courses

Create Course

My Resources

Upload Resource

My Bundles

Create Bundle

Demo Mode - Switch Role:

Student

Tutor

Admin

📖

Total Courses

8

+2 this month

👥

Total Students

342

+15 this week

💰

Total Earnings

\$12,450

+\$1,250 this month

📈

Active Enrollments

89

Recent Reviews

Alice Cooper

React Fundamentals

Excellent teaching style!

2 hours ago

★★★★★

Bob Martinez

JavaScript Mastery

Very helpful and clear explanations.

5 hours ago

★★★★☆

📖

Create Course

📈

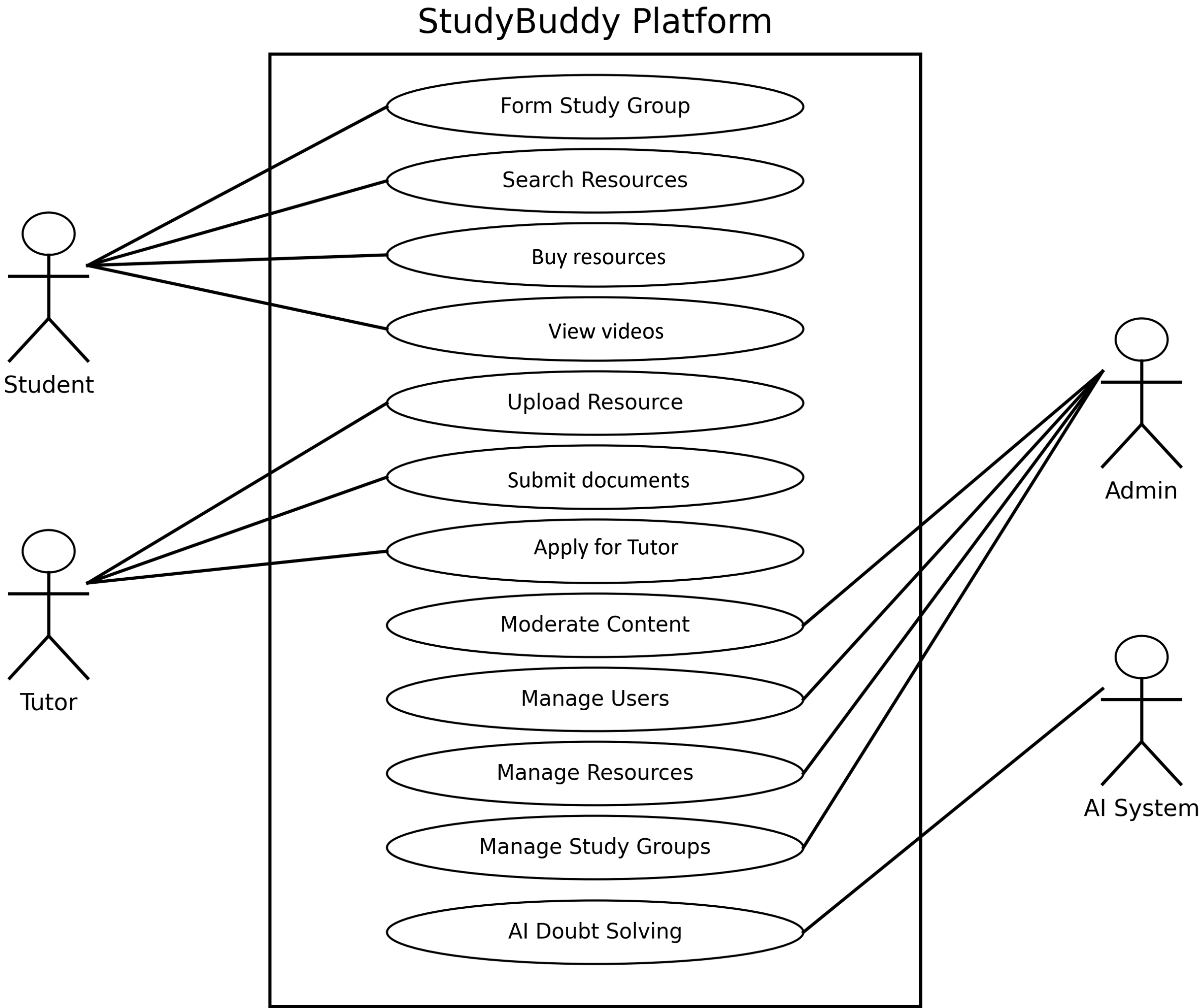
Upload Resource

💰

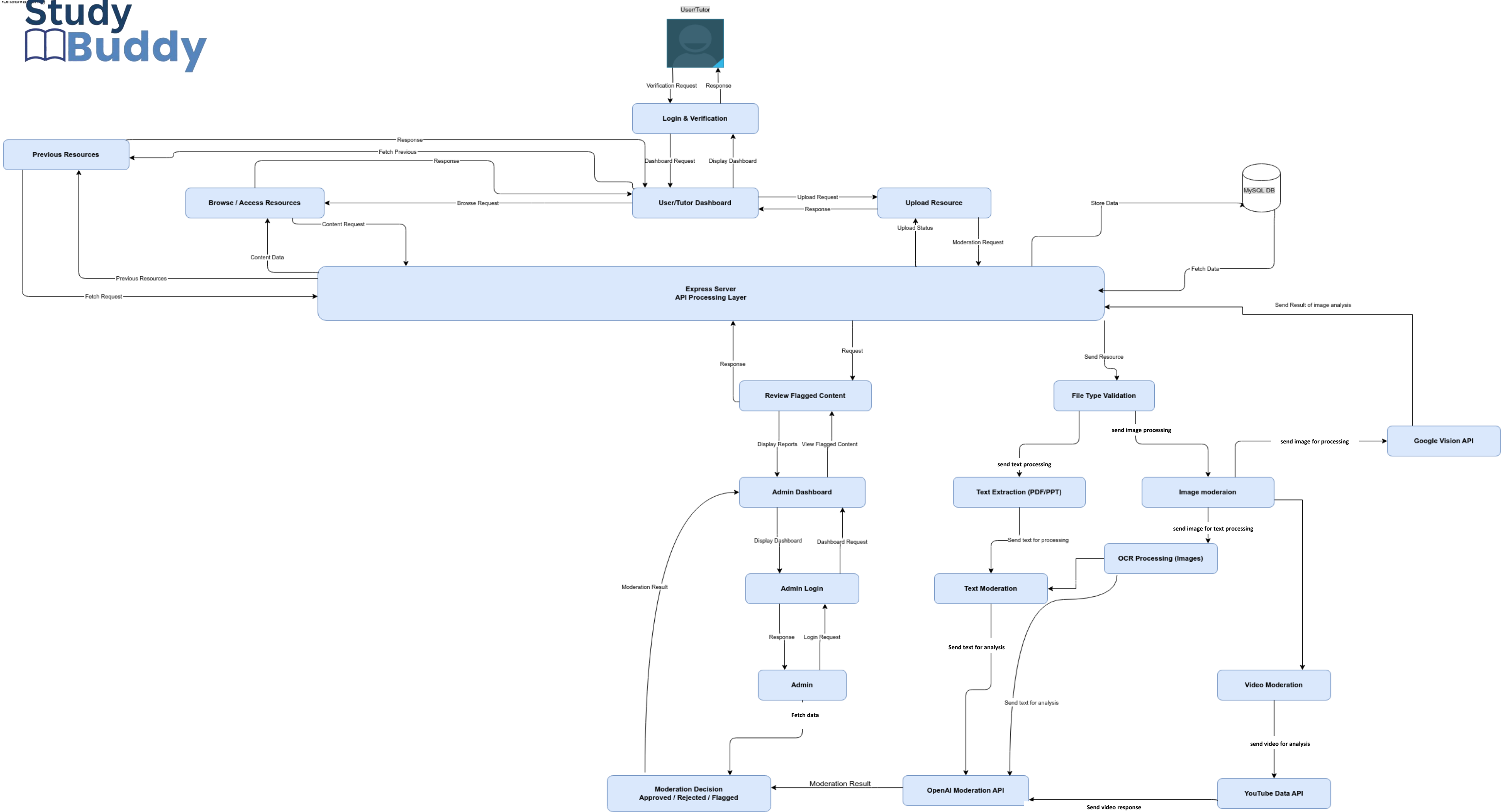
View Earnings

StudyBuddy v1.0

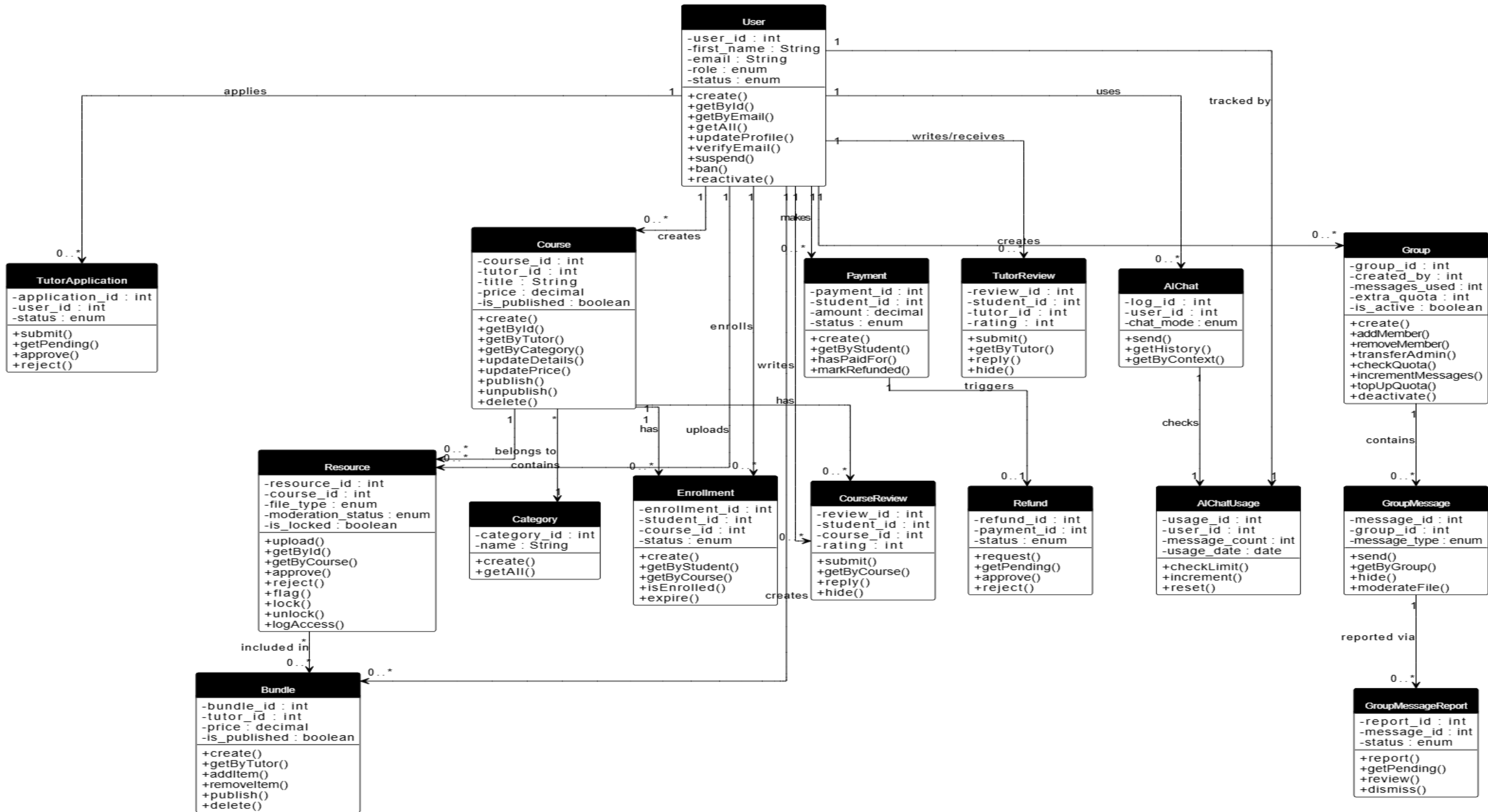
Use case Diagram



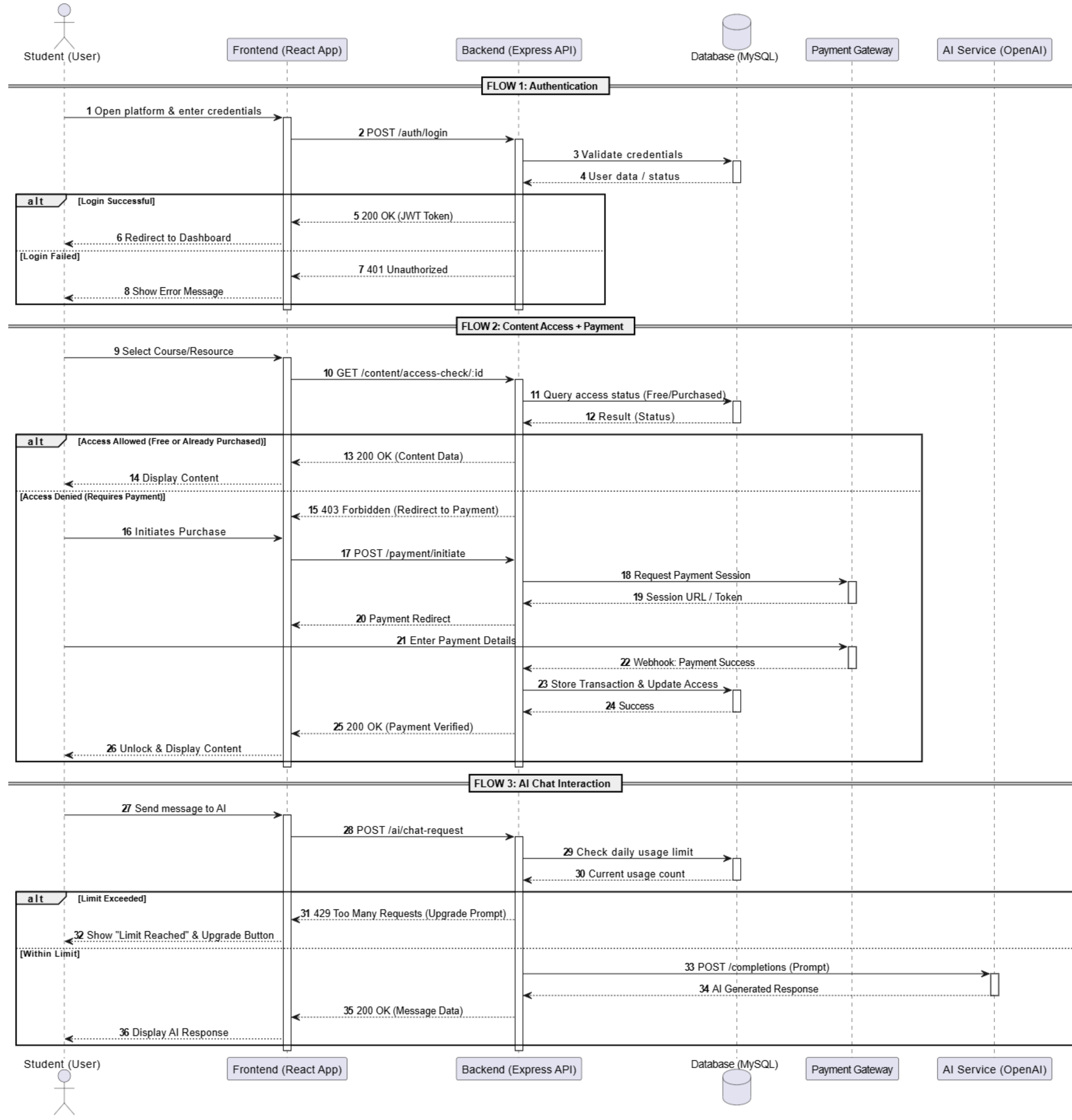
Architecture Diagram

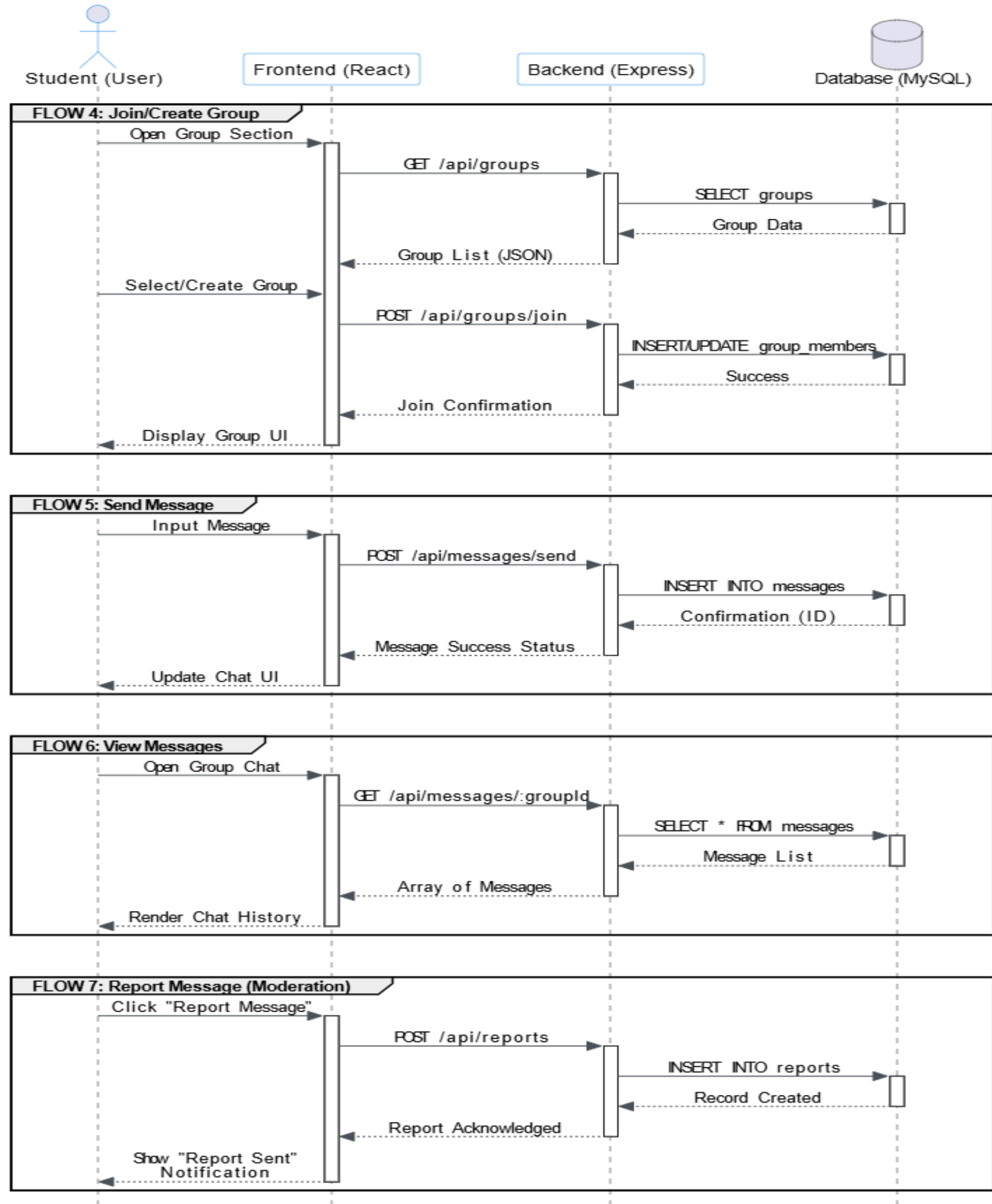


Class Diagram

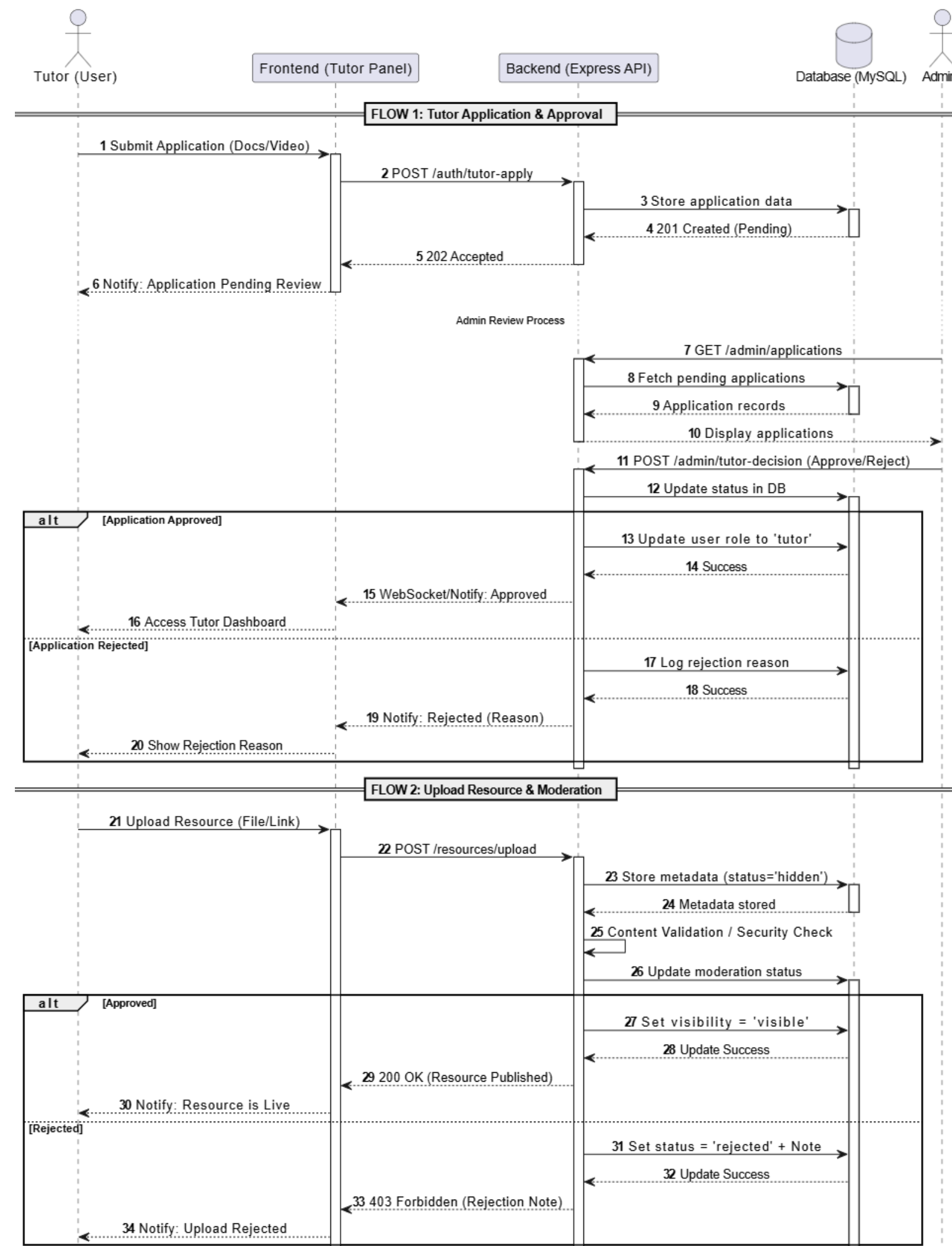


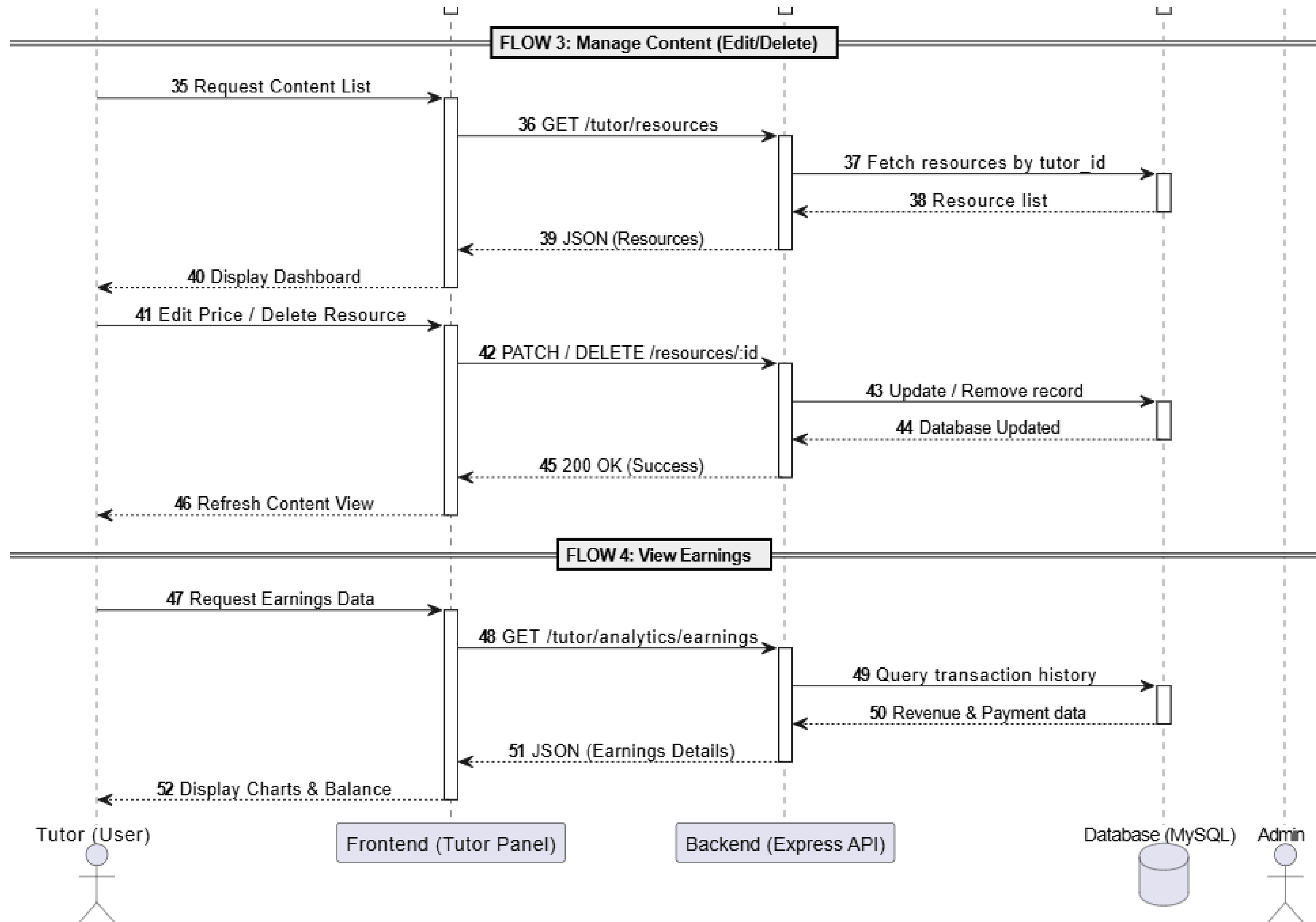
Sequence(User) Diagram



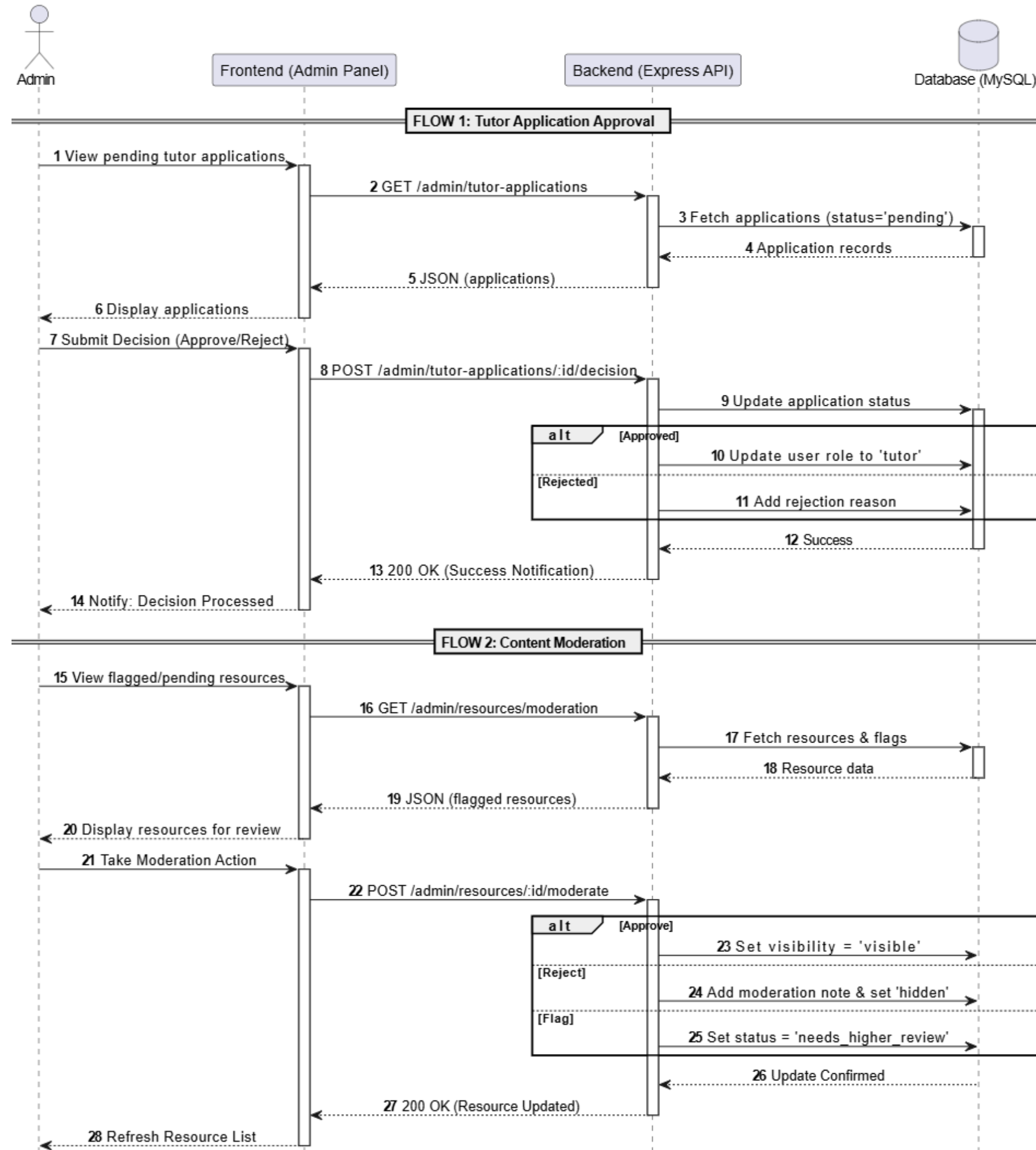


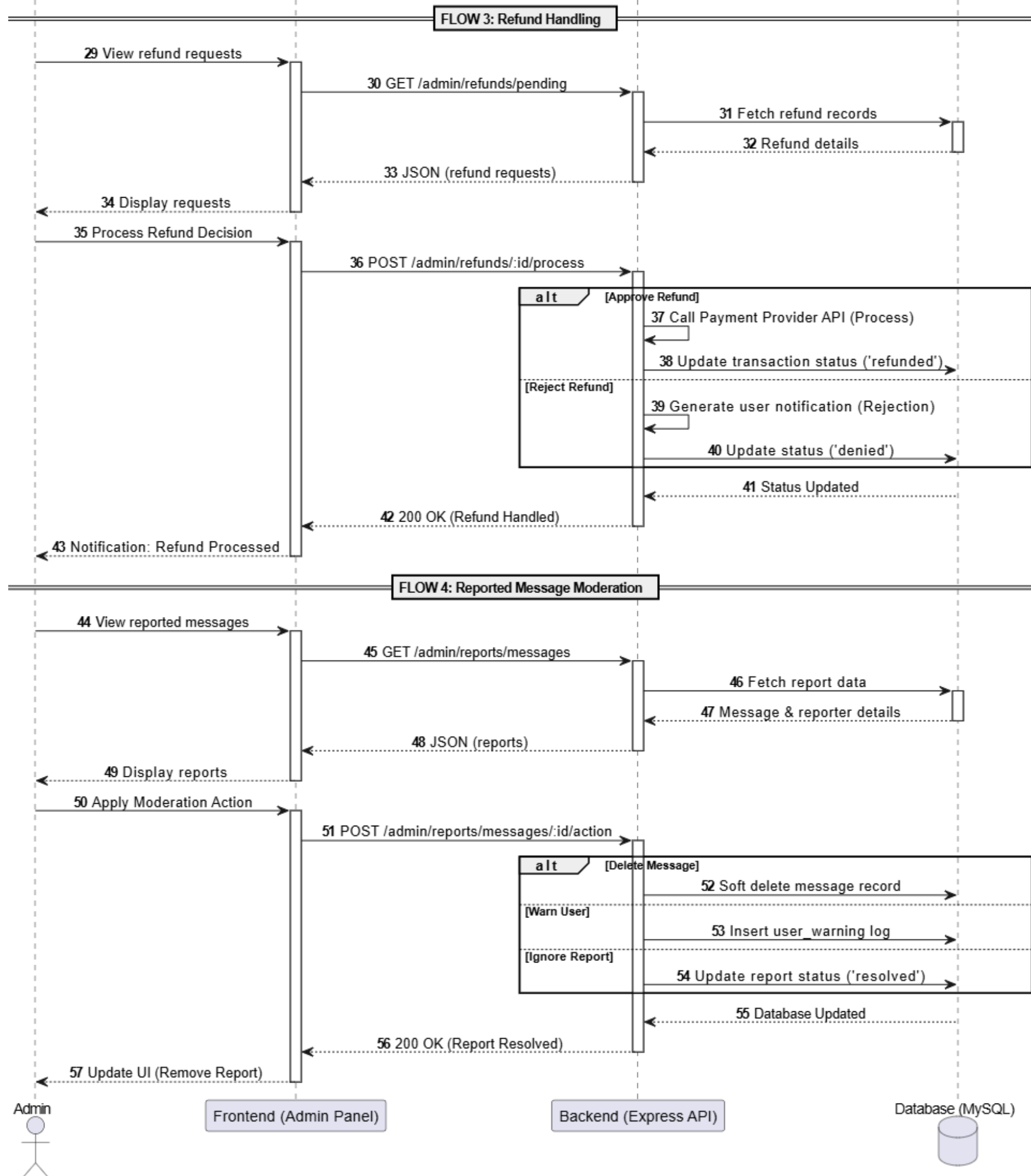
Sequence(Tutor) Diagram



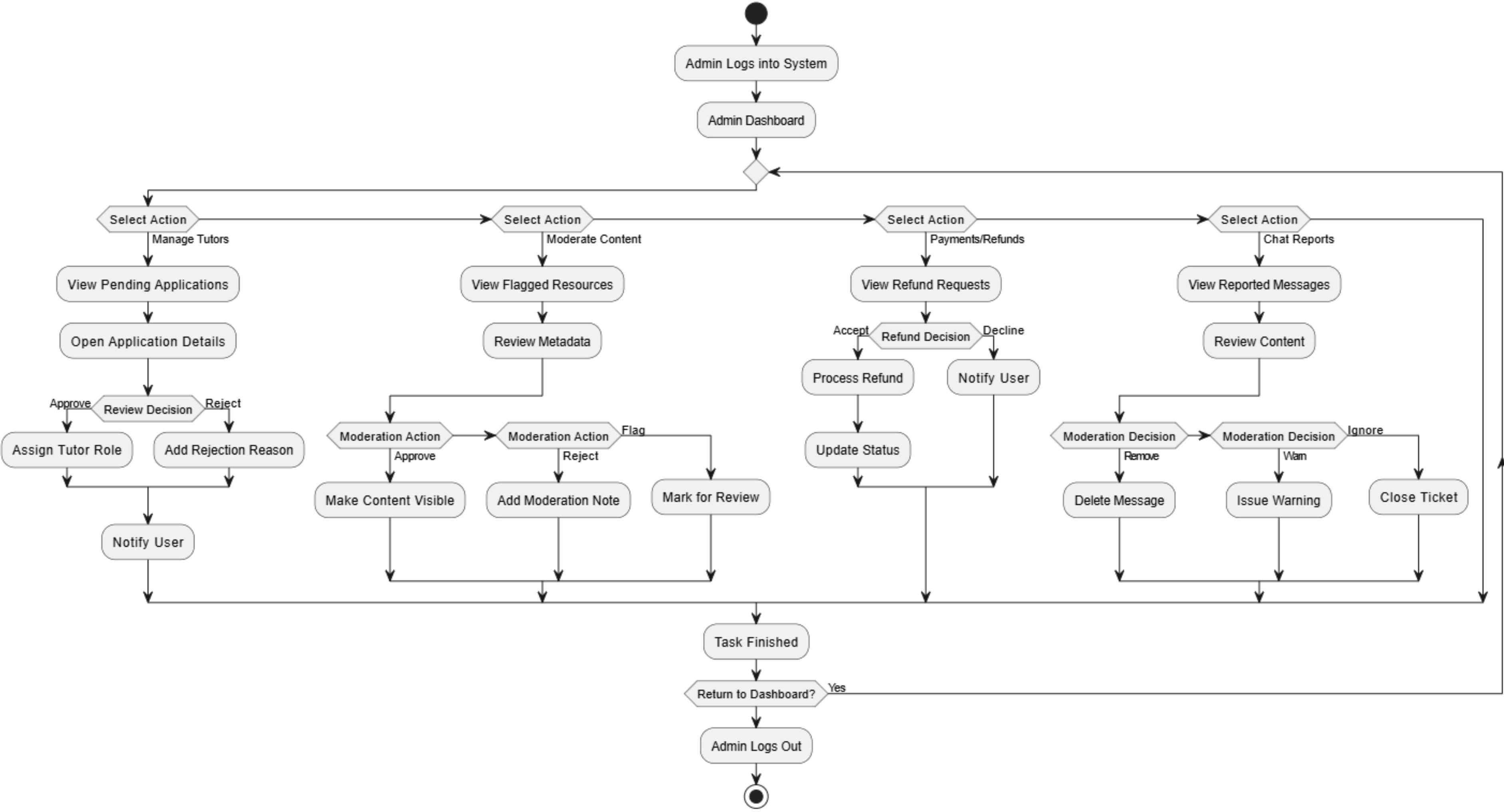


Sequence(Admin) Diagram

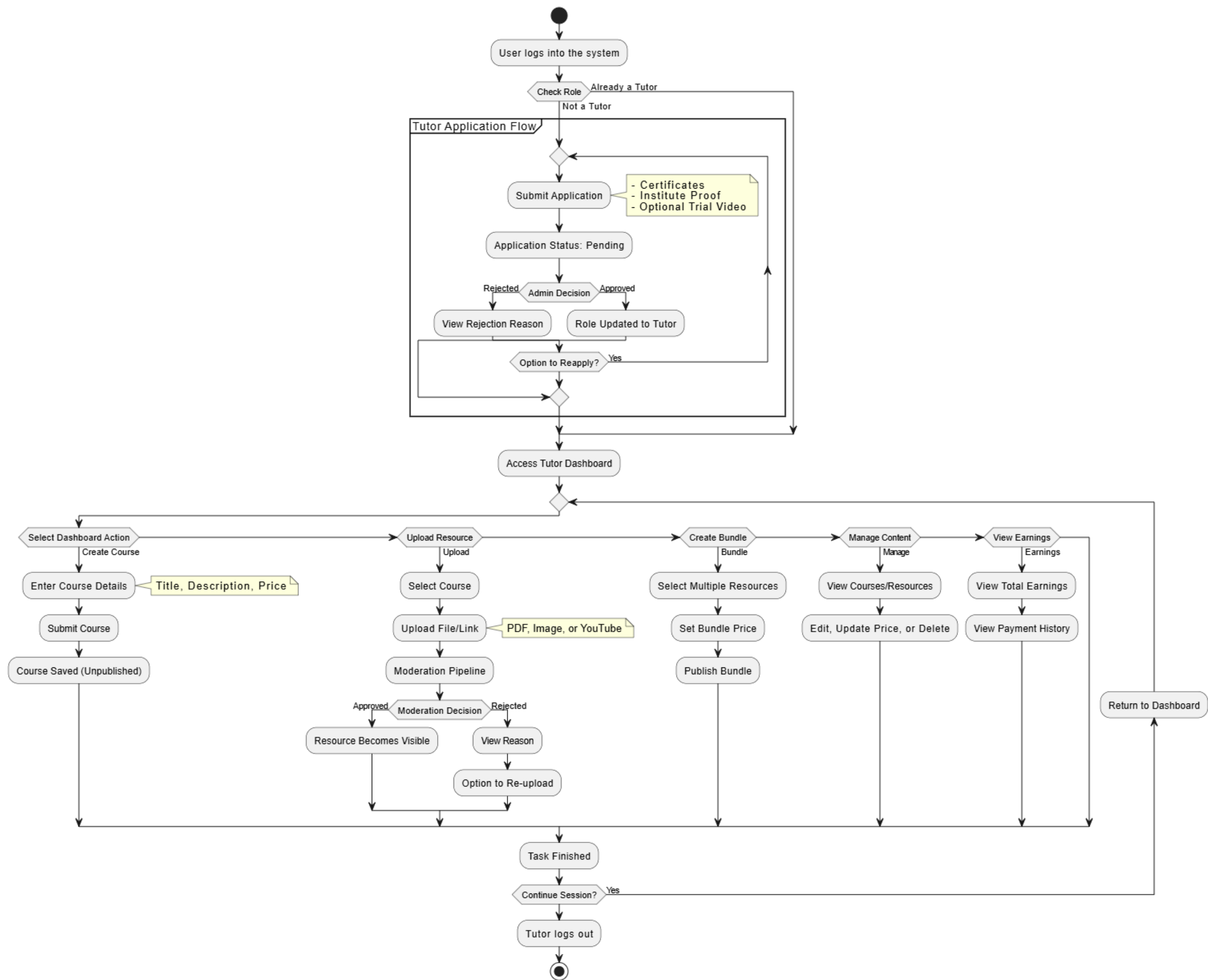




Activity(Admin) Diagram

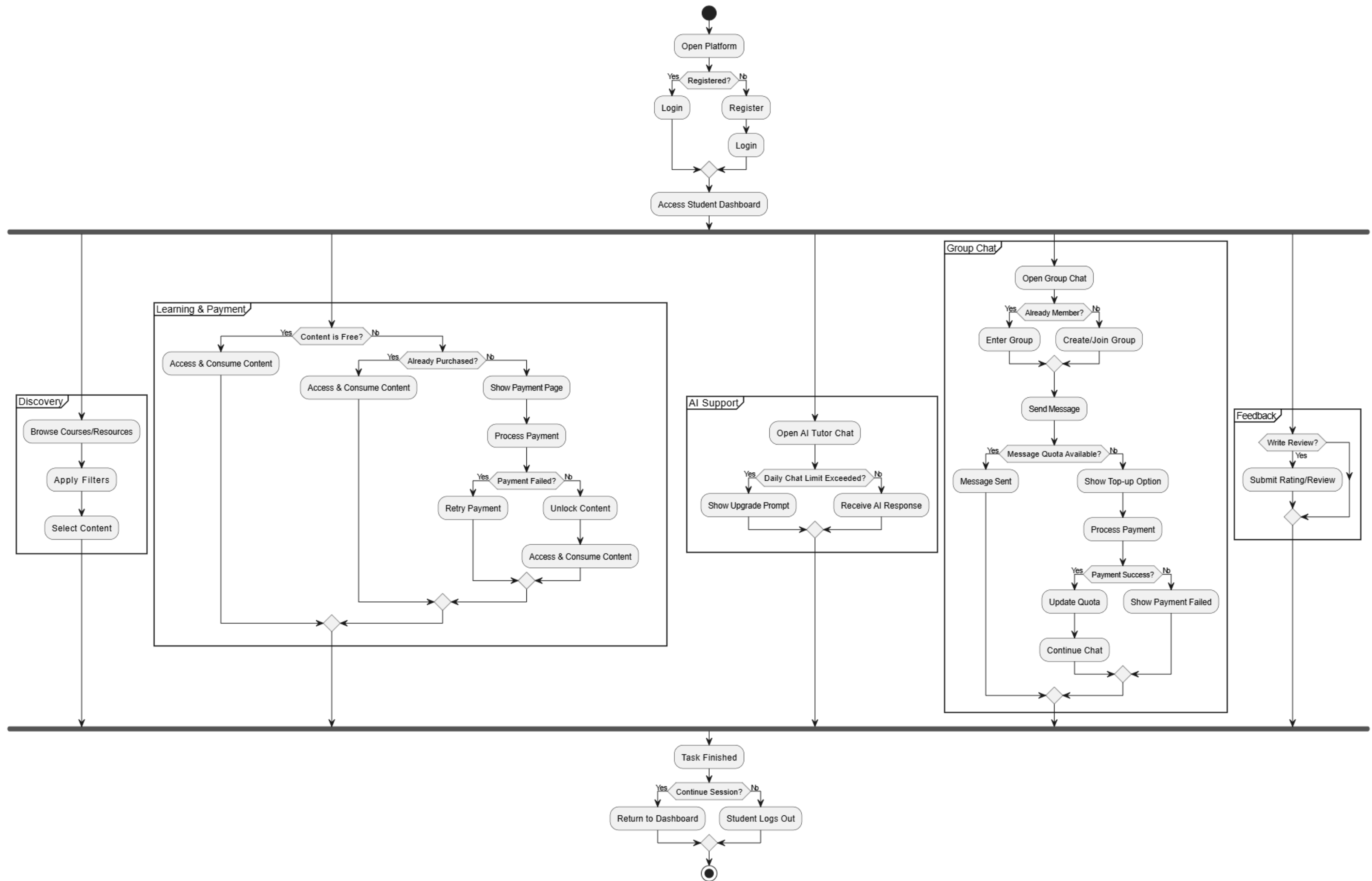


Activity(Tutor) Diagram



Activity(User) Diagram

Student Activity Diagram - StudyBuddy



Data Dictinoary

Table 1: users

Stores all user accounts across all roles (Student, Tutor, Admin).

Field	Data Type	Constraints	Description
<u>user_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique identifier for each user
name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(150)	NOT NULL, UNIQUE	Login email address
<u>password_hash</u>	VARCHAR(255)	NOT NULL	Bcrypt hashed password
role	ENUM('student', 'tutor', 'admin')	NOT NULL, DEFAULT 'student'	Role assigned to the user
status	ENUM('active', 'suspended', 'banned')	DEFAULT 'active'	Account status
<u>profile_pic</u>	VARCHAR(255)	NULL	Profile picture URL/path
<u>is_verified</u>	BOOLEAN	DEFAULT FALSE	Whether email is verified
<u>created_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

Table 2: tutor_profiles

Stores extended profile information specific to tutors.

Field	Data Type	Constraints	Description
<u>profile_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique profile identifier
<u>tutor_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Reference to the tutor user
bio	TEXT	NULL	Short biography of the tutor
<u>institute_name</u>	VARCHAR(200)	NULL	Current or past institution
<u>experience_years</u>	INT	DEFAULT 0	Years of teaching experience
<u>created_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Profile creation timestamp

Table 3: tutor_skills

Stores individual skill tags associated with a tutor (one row per skill).

Field	Data Type	Constraints	Description
<u>skill_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique skill entry ID
<u>tutor_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Tutor who owns this skill
<u>skill_name</u>	VARCHAR(100)	NOT NULL	Name of the skill (e.g., Python, React)
proficiency	ENUM('beginner', 'intermediate', 'expert')	DEFAULT 'intermediate'	Proficiency level

Table 4: tutor_applications

Tracks tutor applications submitted by users seeking tutor role approval.

Field	Data Type	Constraints	Description
<u>application_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique application ID
<u>user_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Applicant user reference
<u>trial_video_url</u>	VARCHAR(500)	NULL	Optional trial teaching video URL
status	ENUM('pending', 'approved', 'rejected')	DEFAULT 'pending'	Application status
<u>admin_note</u>	TEXT	NULL	Admin remarks or rejection reason
<u>reviewed_by</u>	INT	FK → <u>users.user_id</u> , NULL	Admin who reviewed the application
<u>reviewed_at</u>	DATETIME	NULL	Timestamp of admin decision
<u>applied_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Application submission time

Table 5: tutor_application_documents

Stores documents (certificates, institute letters) submitted with a tutor application.

Field	Data Type	Constraints	Description
<u>doc_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique document record ID
<u>application_id</u>	INT	FK → <u>tutor_applications.application_id</u> , NOT NULL	Associated application
<u>document_url</u>	VARCHAR(500)	NOT NULL	URL/path to the uploaded document
<u>document_type</u>	VARCHAR(100)	NULL	Type (e.g., Institute Letter, Certificate)
<u>uploaded_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Upload timestamp

Table 6: categories

Stores subject categories used to classify courses and resources.

Field	Data Type	Constraints	Description
<u>category_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique category identifier
name	VARCHAR(100)	NOT NULL, UNIQUE	Category name (e.g., Programming, Math)
description	TEXT	NULL	Brief description of the category

Table 7: courses

Stores all courses created by tutors on the platform.

Field	Data Type	Constraints	Description
<u>course_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique course identifier
<u>tutor_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Tutor who created the course
<u>category_id</u>	INT	FK → <u>categories.category_id</u> , NULL	Course category
title	VARCHAR(200)	NOT NULL	Title of the course
description	TEXT	NULL	Detailed course description
price	DECIMAL(10,2)	DEFAULT 0.00	Course price (0 = free)
<u>is_published</u>	BOOLEAN	DEFAULT FALSE	Whether the course is live
<u>created_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Course creation timestamp

Table 8: resources

Stores all educational resources uploaded by tutors (PDF, image, YouTube link).

Field	Data Type	Constraints	Description
<u>resource_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique resource identifier
<u>course_id</u>	INT	FK → <u>courses.course_id</u> , NOT NULL	Course the resource belongs to
<u>uploaded_by</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Tutor who uploaded the resource
<u>file_name</u>	VARCHAR(255)	NOT NULL	Original file name
<u>file_type</u>	ENUM('pdf' 'image' 'youtube')	NOT NULL	Type of resource
<u>file_url</u>	VARCHAR(500)	NOT NULL	Storage path or YouTube URL
<u>is_locked</u>	BOOLEAN	DEFAULT FALSE	Whether resource requires payment
price	DECIMAL(10,2)	DEFAULT 0.00	Individual resource price
<u>moderation_status</u>	ENUM('pending' 'approved' 'rejected' 'flagged')	DEFAULT 'pending'	Status after moderation pipeline
<u>moderation_notes</u>	TEXT	NULL	Reason for rejection or flag
<u>created_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Upload timestamp

Table 9: resource_categories

Junction table linking resources to multiple categories (many-to-many).

Field	Data Type	Constraints	Description
resource_id	INT	FK → resources.resource_id, NOT NULL	Resource reference
category_id	INT	FK → categories.category_id, NOT NULL	Category reference

Table 10: resource_moderation_logs

Logs each step of the 6-step automated moderation pipeline for a resource.

Field	Data Type	Constraints	Description
log_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique log entry ID
resource_id	INT	FK → resources.resource_id, NOT NULL	Resource being moderated
step_name	VARCHAR(100)	NOT NULL	Pipeline step name
step_status	ENUM('pass' 'fail' 'skip')	NOT NULL	Result of this step
step_notes	TEXT	NULL	Details from the moderation step
checked_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp of this step

Table 11: resource_extracted_content

Stores text extracted from resources (PDF text, OCR output) used in moderation.

Field	Data Type	Constraints	Description
<u>extraction_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique extraction record
<u>resource_id</u>	INT	FK → <u>resources.resource_id</u> , NOT NULL	Source resource
<u>extracted_text</u>	LONGTEXT	NULL	Full extracted text content
<u>extraction_method</u>	ENUM('pdf_parse', 'ocr', 'youtube_api')	NOT NULL	Method used to extract
<u>extracted_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Extraction timestamp

Table 12: video_metadata

Stores YouTube video metadata fetched during the moderation pipeline.

Field	Data Type	Constraints	Description
<u>meta_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique metadata record
<u>resource_id</u>	INT	FK → <u>resources.resource_id</u> , NOT NULL	Associated resource
<u>youtube_video_id</u>	VARCHAR(50)	NOT NULL	YouTube video ID
title	VARCHAR(500)	NULL	Video title from YouTube API
<u>duration_seconds</u>	INT	NULL	Video duration in seconds
<u>is_embeddable</u>	BOOLEAN	NULL	Whether video can be embedded
<u>fetched_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Fetch timestamp

Table 13: video_moderation_scenes

Stores scene-level moderation results from Google Vision API applied to video thumbnails.

Field	Data Type	Constraints	Description
<u>scene_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique scene record
<u>resource_id</u>	INT	FK → <u>resources_resource_id</u> , NOT NULL	Source resource
<u>scene_timestamp</u>	INT	NULL	Timestamp of the scene (seconds)
<u>violence_score</u>	FLOAT	NULL	Violence likelihood score (0-1)
<u>adult_score</u>	FLOAT	NULL	Adult content likelihood score (0-1)
flagged	BOOLEAN	DEFAULT FALSE	Whether this scene was flagged

Table 14: enrollments

Tracks student enrollments in courses.

Field	Data Type	Constraints	Description
<u>enrollment_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique enrollment record
<u>student_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Enrolled student
<u>course_id</u>	INT	FK → <u>courses.course_id</u> , NOT NULL	Course enrolled in
<u>price_at_enrollment</u>	DECIMAL(10,2)	DEFAULT 0.00	Price paid at time of enrollment
status	ENUM('active', 'expired', 'cancelled')	DEFAULT 'active'	Enrollment status
<u>enrolled_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Enrollment timestamp

Table 15: bundles

Stores curated resource bundles created by tutors.

Field	Data Type	Constraints	Description
<u>bundle_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique bundle identifier
<u>tutor_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Tutor who created the bundle
title	VARCHAR(200)	NOT NULL	Bundle name
description	TEXT	NULL	Bundle description
price	DECIMAL(10,2)	NOT NULL	Bundle price
<u>is_published</u>	BOOLEAN	DEFAULT FALSE	Whether bundle is visible
<u>created_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Bundle creation timestamp

Table 16: bundle_items

Junction table linking bundles to their included resources (one row per resource per bundle).

Field	Data Type	Constraints	Description
<u>bundle_id</u>	INT	FK → <u>bundles.bundle_id</u> , NOT NULL	Bundle reference
<u>resource_id</u>	INT	FK → <u>resources.resource_id</u> , NOT NULL	Resource included in the bundle

Table 17: payments

Tracks all payment transactions on the platform. Exactly one of course_id, resource_id, or bundle_id is populated per transaction.

Field	Data Type	Constraints	Description
<u>payment_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique payment record
<u>student_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Student who made payment
<u>course_id</u>	INT	FK → <u>courses.course_id</u> , NULL	Populated if buying a course
<u>resource_id</u>	INT	FK → <u>resources.resource_id</u> , NULL	Populated if buying a resource
<u>bundle_id</u>	INT	FK → <u>bundles.bundle_id</u> , NULL	Populated if buying a bundle
amount	DECIMAL(10,2)	NOT NULL	Amount paid
status	ENUM('success', 'failed', 'pending')	NOT NULL	Payment status
<u>paid_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Payment timestamp

Table 18: refunds

Tracks refund requests and their processing status.

Field	Data Type	Constraints	Description
<u>refund_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique refund record
<u>payment_id</u>	INT	FK → <u>payments.payment_id</u> , NOT NULL	Original payment being refunded
<u>student_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Student requesting refund
reason	TEXT	NULL	Reason provided for the refund request
status	ENUM('pending','approved','rejected')	DEFAULT 'pending'	Refund status
<u>reviewed_by</u>	INT	FK → <u>users.user_id</u> , NULL	Admin who processed the refund
<u>reviewed_at</u>	DATETIME	NULL	Timestamp of refund decision
<u>requested_at</u>	DATETIME	DEFAULT , CURRENT_TIMESTAMP	Refund request timestamp

Table 19: course_reviews

Stores student reviews and ratings for courses they have enrolled in.

Field	Data Type	Constraints	Description
<u>review_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique review identifier
<u>student_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Student who wrote the review
<u>course_id</u>	INT	FK → <u>courses.course_id</u> , NOT NULL	Course being reviewed
rating	TINYINT	NOT NULL (1-5)	Star rating (1 to 5)
comment	TEXT	NULL	Written review text
<u>created_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Review submission timestamp

Table 20: tutor_reviews

Stores student reviews and ratings for tutors.

Field	Data Type	Constraints	Description
<u>review_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique review identifier
<u>student_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Student who wrote the review
<u>tutor_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Tutor being reviewed
rating	TINYINT	NOT NULL (1-5)	Star rating (1 to 5)
comment	TEXT	NULL	Written review text
<u>created_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Review submission timestamp

Table 21: ai_chat_logs

Logs all AI chatbot conversations between students and the OpenAI-powered assistant.

Field	Data Type	Constraints	Description
log_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique log entry ID
student_id	INT	FK → users.user_id, NOT NULL	Student using the AI chat
course_id	INT	FK → courses.course_id, NULL	Course context if applicable
message	TEXT	NOT NULL	Student's message/query
response	TEXT	NOT NULL	AI-generated response
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Chat message timestamp

Table 22: ai_chat_usage

Tracks each student's daily AI chat usage for enforcing the per-day message limit.

Field	Data Type	Constraints	Description
usage_id	INT	PK, NOT NULL, AUTO_INCREMENT	Unique usage record
student_id	INT	FK → users.user_id, NOT NULL	Student being tracked
chats_used	INT	DEFAULT 0	Number of AI chats used today
chat_limit	INT	DEFAULT 10	Daily limit for this student
reset_date	DATE	NOT NULL	Date when the count was last reset

Table 23: groups

Stores learning groups created by students or tutors.

Field	Data Type	Constraints	Description
<u>group_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique group identifier
name	VARCHAR(150)	NOT NULL	Group name
<u>created_by</u>	INT	FK → <u>users.user_id</u> , NOT NULL	User who created the group
<u>course_id</u>	INT	FK → <u>courses.course_id</u> , NULL	Associated course (if any)
<u>message_quota</u>	INT	DEFAULT 100	Shared message quota for the group
<u>messages_used</u>	INT	DEFAULT 0	Messages consumed so far
<u>created_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Group creation timestamp

Table 24: group_members

Tracks membership of users in groups.

Field	Data Type	Constraints	Description
<u>member_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique membership record
<u>group_id</u>	INT	FK → <u>groups.group_id</u> , NOT NULL	Group reference
<u>user_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Member user reference
role	ENUM('member' 'admin')	DEFAULT 'member'	Member's role in the group
<u>joined_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Membership join timestamp

Table 25: group_messages

Stores messages sent within groups.

Field	Data Type	Constraints	Description
<u>message_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique message identifier
<u>group_id</u>	INT	FK → <u>groups_group_id</u> , NOT NULL	Group the message belongs to
<u>sender_id</u>	INT	FK → <u>users_user_id</u> , NOT NULL	User who sent the message
content	TEXT	NOT NULL	Message text content
<u>is_deleted</u>	BOOLEAN	DEFAULT FALSE	Soft delete flag
<u>sent_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Message send timestamp

Table 26: group_message_reports

Stores reports submitted by users for inappropriate group messages.

Field	Data Type	Constraints	Description
<u>report_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique report record
<u>message_id</u>	INT	FK → <u>group_messages_message_id</u> , NOT NULL	Reported message
<u>reported_by</u>	INT	FK → <u>users_user_id</u> , NOT NULL	User who submitted the report
reason	TEXT	NULL	Reason for the report
status	ENUM('pending','reviewed','dismissed')	DEFAULT 'pending'	Admin review status
<u>reviewed_by</u>	INT	FK → <u>users_user_id</u> , NULL	Admin who reviewed the report
<u>reviewed_at</u>	DATETIME	NULL	Timestamp of admin review
<u>reported_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Report submission timestamp

Table 27: group_payments

Tracks payments made by group members to top up the group's shared message quota.

Field	Data Type	Constraints	Description
<u>gpay_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique group payment record
<u>group_id</u>	INT	FK → <u>groups.group_id</u> , NOT NULL	Group receiving the top-up
<u>paid_by</u>	INT	FK → <u>users.user_id</u> , NOT NULL	User who made the payment
amount	DECIMAL(10,2)	NOT NULL	Amount paid
<u>quota_added</u>	INT	NOT NULL	Additional message quota granted
<u>paid_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Payment timestamp

Table 28: resource_access_logs

Note: This table was evaluated during design and dropped. Free standalone resources do not support reviews, and course resources use enrollment records as proof of access. Paid resources use the payments table. No extra access log table is required.

Field	Data Type	Constraints	Description
<u>log_id</u>	INT	PK, NOT NULL, AUTO_INCREMENT	Unique log entry ID
<u>student_id</u>	INT	FK → <u>users.user_id</u> , NOT NULL	Student who accessed the resource
<u>resource_id</u>	INT	FK → <u>resources.resource_id</u> , NOT NULL	Resource that was accessed
<u>course_id</u>	INT	FK → <u>courses.course_id</u> , NULL	Course context if resource belongs to a course
<u>accessed_at</u>	DATETIME	DEFAULT CURRENT_TIMESTAMP	Timestamp of access

MIS Reports design

Report 1: User Registration Report

Report ID: RPT-USR-001 | For: Admin | Frequency: Weekly / Monthly

This report provides a summary of all new user registrations over a specified time period, broken down by role and account status. It helps the admin monitor platform growth and track how many students and tutors are joining the platform.

Registration Date	Full Name	Email	Role	Status
01/04/2025	Riya Sharma	riya@gmail.com	Student	Active
02/04/2025	Arjun Mehta	arjun@gmail.com	Student	Active
03/04/2025	Priya Patel	priya@gmail.com	Tutor	Active
04/04/2025	Karan Desai	karan@gmail.com	Student	Suspended
05/04/2025	Neha Joshi	neha@gmail.com	Tutor	Active
TOTAL: 5 Users	—	—	3 Students, 2 Tutors	4 Active

Filters: Date range, Role, Status

Report 2: Tutor Verification Status Report

Report ID: RPT-TUT-001 | For: Admin | Frequency: On demand

This report tracks all tutor applications submitted to the platform along with their current approval status. It allows the admin to monitor pending verifications, view rejection reasons, and ensure that only credentialed educators are approved to publish content.

Applicant Name	Documents	Trial Video	Status	Applied Date	Review Date
Priya Patel	2	Yes	Approved	03/04/2025	05/04/2025
Neha Joshi	1	No	Pending	05/04/2025	—
Amit Shah	3	Yes	Rejected	01/04/2025	04/04/2025
Sneha Verma	2	Yes	Approved	28/03/2025	02/04/2025
TOTAL: 4	—	—	2 Approved, 1 Pending, 1 Rejected	—	—

Filters: Status, Date range

Report 3: Course Enrollment Report

Report ID: RPT-CRS-001 | For: Admin / Tutor | Frequency: Weekly / Monthly

This report summarises enrollment activity across all published courses. It helps tutors and the admin understand which courses are most popular, how many students are paying versus enrolling for free, and how much revenue each course is generating.

Course Title	Tutor Name	Total Enrollments	Paid Enrollments	Revenue (₹)	Published
Python Basics	Priya Patel	42	30	8,970	01/03/2025
Web Dev with React	Sneha Verma	35	35	17,465	15/02/2025
Data Structures	Priya Patel	28	20	4,980	10/03/2025
MySQL for Beginners	Sneha Verma	19	10	2,490	20/03/2025
TOTAL	—	124	95	33,905	—

Filters: Tutor, Category, Date range

Report 4: Payment & Revenue Report

Report ID: RPT-PAY-001 | For: Admin | Frequency: Daily / Monthly

This report tracks all payment transactions made on the platform across courses, individual resources, and bundles. It gives the admin a complete picture of revenue flow, payment success rates, and failed or pending transactions that may need follow-up.

Payment ID	Student Name	Purchase Type	Item Name	Amount (₹)	Status	Date
PAY001	Riya Sharma	Course	Python Basics	299	Success	01/04/2025
PAY002	Arjun Mehta	Bundle	Dev Starter Pack	499	Success	02/04/2025
PAY003	Karan Desai	Resource	React Cheatsheet PDF	99	Failed	03/04/2025
PAY004	Meera Nair	Course	Web Dev with React	499	Success	04/04/2025
PAY005	Rohan Gupta	Resource	MySQL Notes PDF	49	Pending	05/04/2025
TOTAL: 5	—	—	—	1,445	3 Success	—

Filters: Purchase type, Status, Date range

Report 5: Content Moderation Report

Report ID: RPT-MOD-001 | For: Admin | Frequency: Daily / On demand

This report lists all resources that have been processed through the automated 6-step content moderation pipeline. It highlights resources that were flagged or rejected, along with the specific pipeline step that failed, helping the admin take corrective action efficiently.

Resource Title	Type	Tutor Name	Step Failed	Status	Upload Date
Python Notes Ch1	PDF	Priya Patel	—	Approved	01/04/2025
React Components	Image	Sneha Verma	Google Vision API	Flagged	02/04/2025
SQL Tutorial Video	YouTube	Priya Patel	YouTube API Check	Rejected	03/04/2025
Data Structures PDF	PDF	Sneha Verma	—	Approved	04/04/2025
CSS Tricks Image	Image	Amit Shah	OpenAI Text Mod.	Flagged	05/04/2025
TOTAL: 5	—	—	3 Issues Found	2 Approved, 2 Flagged, 1 Rejected	—

Filters: Status, Resource type, Date range, Tutor

Report 6: Tutor Performance Report

Report ID: RPT-TUT-002 | For: Admin / Tutor | Frequency: Monthly

This report provides a performance summary for each approved tutor on the platform. It covers the number of courses created, total students taught, average ratings received, and total earnings — giving both the admin and individual tutors a clear view of their impact and revenue.

Tutor Name	Total Courses	Total Students	Avg. Rating	Total Reviews	Earnings (₹)
Priya Patel	3	70	4.7	38	13,950
Sneha Verma	2	54	4.5	29	19,955
Amit Shah	1	0	—	0	0
TOTAL	6	124	4.6 Avg	67	33,905

Filters: Date range, Rating range

Report 7: Refund Report

Report ID: RPT-REF-001 | For: Admin | Frequency: On demand / Monthly

This report tracks all refund requests submitted by students, including the original purchase details, reason for refund, current status, and resolution timeline. It helps the admin process refunds efficiently and identify patterns in refund requests across courses, resources, and bundles.

Student Name	Item Purchased	Amount (₹)	Reason	Status	Request Date	Resolved
Karan Desai	Python Basics	299	Content not as described	Approved	03/04/2025	05/04/2025
Rohan Gupta	Dev Starter Pack	499	Duplicate purchase	Pending	05/04/2025	—
Aisha Khan	React Cheatsheet	99	File not opening	Rejected	02/04/2025	04/04/2025
TOTAL: 3	—	897	—	1 Approved, 1 Pending, 1 Rejected	—	—

Filters: Status, Date range, Amount range

Conclusion

Ed-Tech Platform — Mini Project I Summary

Three-Role Architecture

Student, Tutor & Admin ensure a secure, structured learning environment.

AI Doubt Resolution

OpenAI-powered Q&A with daily usage limits for cost-controlled help.

Scalable DB Design

28 normalized tables across 11 groups for integrity & scalability.

Tutor Verification

Only credentialed educators can publish, maintaining content quality.

Group Chat & Collaboration

Peer learning via group chat with full admin oversight.

Core Skills Demonstrated

Web dev, DB design, RESTful APIs & third-party API integrations.

Automated Moderation

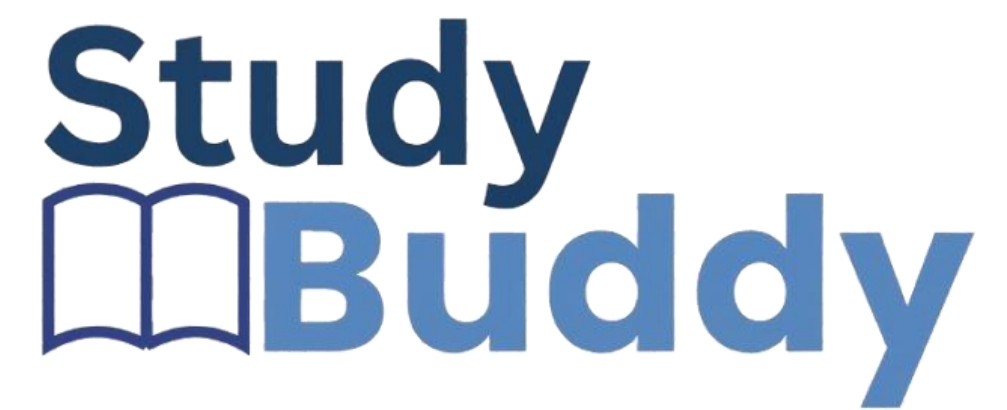
6-step pipeline reduces admin workload & keeps resources appropriate.

Flexible Monetization

Courses, bundles & individual resources for tutors and students.

Future Scope

Foundation laid for live tutoring & mobile app development.



Thank You